

## Interacting Multipoles of Rank 1 and 4:

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_\alpha\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{105} \cdot \left[ \begin{array}{l} -945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \delta(\delta, \epsilon) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \delta(\gamma, \epsilon) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \delta(\gamma, \delta) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \delta(\beta, \epsilon) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\beta, \delta) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \delta(\beta, \gamma) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \epsilon) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\alpha, \delta) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \delta(\alpha, \gamma) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \delta(\alpha, \beta) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \gamma) \cdot \delta(\delta, \epsilon) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \delta) \cdot \delta(\gamma, \epsilon) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \epsilon) \cdot \delta(\gamma, \delta) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \gamma) \cdot \delta(\delta, \epsilon) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \delta) \cdot \delta(\gamma, \epsilon) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \epsilon) \cdot \delta(\gamma, \delta) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \beta) \cdot \delta(\delta, \epsilon) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \epsilon) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \epsilon) \cdot \delta(\beta, \delta) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \epsilon) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \epsilon) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \epsilon) \cdot \delta(\beta, \gamma) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \delta) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \delta) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \gamma) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \end{array} \right] \quad (1)$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_\alpha\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{105} \cdot \left[ \begin{aligned} &-945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\epsilon} \\ &+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \delta(\delta, \delta) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\delta\delta} \\ &+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \delta(\gamma, \gamma) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\gamma\delta} \\ &+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \delta(\gamma, \gamma) \cdot \mu_\alpha \cdot \Phi_{\beta\gamma\gamma\epsilon} \\ &+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \delta(\beta, \beta) \cdot \mu_\alpha \cdot \Phi_{\beta\beta\gamma\delta} \\ &+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\beta, \beta) \cdot \mu_\alpha \cdot \Phi_{\beta\beta\gamma\epsilon} \\ &+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \delta(\beta, \beta) \cdot \mu_\alpha \cdot \Phi_{\beta\beta\delta\epsilon} \\ &+105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\gamma\delta} \\ &+105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\gamma\epsilon} \\ &+105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\delta\epsilon} \\ &+105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \mu_\alpha \cdot \Phi_{\alpha\gamma\delta\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \beta) \cdot \delta(\delta, \delta) \cdot \mu_\alpha \cdot \Phi_{\beta\beta\delta\delta} \\ &-15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \beta) \cdot \delta(\gamma, \gamma) \cdot \mu_\alpha \cdot \Phi_{\beta\beta\gamma\gamma} \\ &-15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \beta) \cdot \delta(\gamma, \gamma) \cdot \mu_\alpha \cdot \Phi_{\beta\beta\gamma\gamma} \\ &-15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \alpha) \cdot \delta(\delta, \delta) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\delta\delta} \\ &-15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\gamma\gamma} \\ &-15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\gamma\gamma} \\ &-15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \alpha) \cdot \delta(\delta, \delta) \cdot \mu_\alpha \cdot \Phi_{\alpha\gamma\delta\delta} \\ &-15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\gamma} \\ &-15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\gamma} \\ &-15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \mu_\alpha \cdot \Phi_{\alpha\gamma\gamma\delta} \\ &-15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\delta} \\ &-15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\delta} \\ &-15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \mu_\alpha \cdot \Phi_{\alpha\gamma\gamma\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_\alpha \cdot \Phi_{\alpha\beta\beta\epsilon} \end{aligned} \right] \quad (2a)$$



Simplify R:

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_\alpha\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{105} \cdot \left[ \begin{array}{l} -945 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Phi_{zzzzz} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Phi_{zz\delta\delta} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Phi_{zz\gamma\gamma} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Phi_{zz\gamma\gamma} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Phi_{zz\beta\beta} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Phi_{zz\beta\beta} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Phi_{zz\beta\beta} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_\alpha \cdot \Phi_{zzz\alpha} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_\alpha \cdot \Phi_{zzz\alpha} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_\alpha \cdot \Phi_{zzz\alpha} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_\alpha \cdot \Phi_{zzz\alpha} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_z \cdot \Phi_{\beta\beta\delta\delta} \\ -30 \cdot R_z^4 \cdot R_z \cdot \mu_z \cdot \Phi_{\beta\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_\alpha \cdot \Phi_{z\alpha\delta\delta} \\ -30 \cdot R_z^4 \cdot R_z \cdot \mu_\alpha \cdot \Phi_{z\alpha\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_\alpha \cdot \Phi_{z\alpha\delta\delta} \\ -30 \cdot R_z^4 \cdot R_z \cdot \mu_\alpha \cdot \Phi_{z\alpha\beta\beta} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_\alpha \cdot \Phi_{z\alpha\gamma\gamma} \\ -30 \cdot R_z^4 \cdot R_z \cdot \mu_\alpha \cdot \Phi_{z\alpha\beta\beta} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_\alpha \cdot \Phi_{z\alpha\gamma\gamma} \\ -30 \cdot R_z^4 \cdot R_z \cdot \mu_\alpha \cdot \Phi_{z\alpha\beta\beta} \end{array} \right] \quad (3a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-11} \cdot -\frac{1}{105} \cdot \left[ \begin{array}{l} +105 \cdot R_z^5 \cdot \mu_\alpha \cdot \Phi_{zzz\alpha} \\ +105 \cdot R_z^5 \cdot \mu_\alpha \cdot \Phi_{zzz\alpha} \\ +105 \cdot R_z^5 \cdot \mu_\alpha \cdot \Phi_{zzz\alpha} \\ +105 \cdot R_z^5 \cdot \mu_\alpha \cdot \Phi_{zzz\alpha} \\ -15 \cdot R_z^5 \cdot \mu_\alpha \cdot \Phi_{z\alpha\delta\delta} \\ -30 \cdot R_z^5 \cdot \mu_\alpha \cdot \Phi_{z\alpha\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_\alpha \cdot \Phi_{z\alpha\delta\delta} \\ -30 \cdot R_z^5 \cdot \mu_\alpha \cdot \Phi_{z\alpha\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_\alpha \cdot \Phi_{z\alpha\gamma\gamma} \\ -30 \cdot R_z^5 \cdot \mu_\alpha \cdot \Phi_{z\alpha\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_\alpha \cdot \Phi_{z\alpha\gamma\gamma} \\ -30 \cdot R_z^5 \cdot \mu_\alpha \cdot \Phi_{z\alpha\beta\beta} \end{array} \right] \quad (3b)$$

$$\xrightarrow{\text{added up}} R_z^{-11} \cdot -\frac{1}{105} \cdot \left[ \begin{array}{l} +420 \cdot R_z^5 \cdot \mu_\alpha \cdot \Phi_{zzz\alpha} \\ -30 \cdot R_z^5 \cdot \mu_\alpha \cdot \Phi_{z\alpha\delta\delta} \\ -60 \cdot R_z^5 \cdot \mu_\alpha \cdot \Phi_{z\alpha\gamma\gamma} \\ -90 \cdot R_z^5 \cdot \mu_\alpha \cdot \Phi_{z\alpha\beta\beta} \end{array} \right] \quad (3c)$$

Simplify Dipoles:

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_\alpha\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{105} \cdot \left[ \begin{array}{l} +420 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yzzz} \\ -30 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -60 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -90 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \end{array} \right] \quad (4a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-11} \cdot -\frac{1}{105} \cdot [+0] \quad (4b)$$

**Combining everything:**

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_{\alpha}\Phi_{\beta\gamma\delta\epsilon} = +0 \tag{5}$$